

Association between histone deacetylase compartmentalization and survival of Retinal Ganglion Cells and Primary Hippocampal Neuron

Author: Malwina Lisek

Class IIa histone deacetylases (HDAC4 and HDAC5) are critical transcriptional regulators whose function depends on intracellular localization. Whether class IIa HDACs are neuroprotective or proapoptotic is controversial. Class IIa HDACs when localized to the nucleus bind MEF2 on chromatin and mediate gene repression. In this study, it was tested if nuclear or cytoplasmic localization of HDAC4/5 and inhibition of MEF2A activity will promote neuronal survival and/or axon outgrowth in primary neurons. Neuronal survival and axon growth were studied using primary cultures of rat E18 hippocampal neurons. Cells were transfected with plasmids encoding wild-type (WT) forms and mutants of HDAC4 and HDAC5 with constitutive nuclear (3SA, S/A) or cytoplasmic (L175A, ANLS) localization. Surviving neurons spontaneously exhibit neurite outgrowth, and measurement of the longest neurite represents an assay for axon growth as a model. With the AAV2 intravitreal injections we want to confirm the neuroprotective effect of HDAC4 and 5 nuclear localization in RGC cells after optic nerve crush.

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